EXP-CZLIFT-f5e06c: evidence review

Decision report for DEC-20261023-9de83a, hypothesis H-CZLIFT-bd2017, goal GOAL-CZLIFT-1516d5

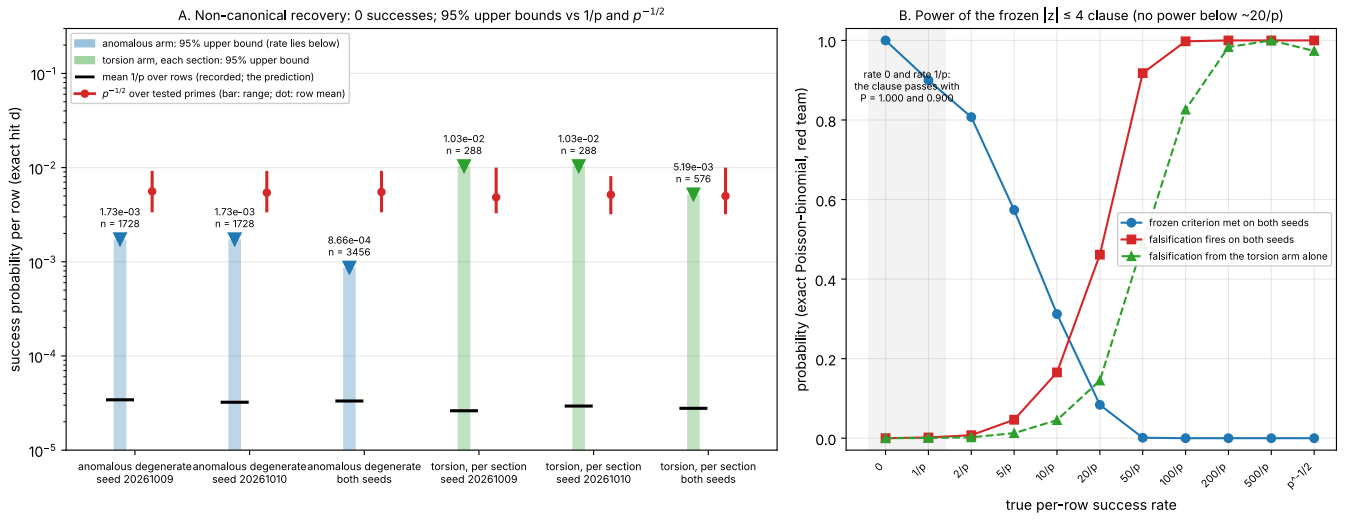
Decision	`refine`. H-CZLIFT-bd2017 moves approved → analyzed (DEC-20261023-9de83a). H-CZLIFT-889c9d and H-CZLIFT-6d4ed1 are not moved here; separate review tasks decide them.
Evidence	EV-CZLIFT-e9a7cf, direction `supports` for the exact clauses only
Strength	`replicated` for the exact clauses (torsion clauses same-code); `inconclusive` for the 1/p clauses; tier clause not met as posed
Claim tier	`toy`. The largest field is $\mathbb{F}_p$ with $p = 97813 < 2^{17}$. The specification's "medium" label is a mislabel.
Runs	RUN-CZLIFT-032a92 (seed 20261009), RUN-CZLIFT-6447f0 (seed 20261010)
Tasks	producer TASK-20261009-951dfd; snapshot TASK-20261009-046d5b; run archive TASK-20261009-4ac56e; validation TASK-20261022-65e6c1; red team TASK-20261023-e5df50; ledger archive TASK-20261023-4d7449
Knowledge	nothing promoted (the decision is refine; see the decision record's `knowledge_promotion`)

Exact scoped claim

At toy tier, the largest field being $\mathbb{F}_p$ with $p = 97813 < 2^{17}$. The data are two clean runs (RUN-CZLIFT-032a92 and RUN-CZLIFT-6447f0) of the unmodified lane-1 drivers over a new sampler with $10^4 \le p \le 10^5$.

- **Anomalous arm.** 41 distinct class-number-one CM anomalous curves ($D \in \{-3, -11, -19, -43, -67, -163\}$) at precision p^8. The j-distance degeneracy rule agreed on all 4608 rows (164 curve × lift-class cells). Smart's ratio equalled d on all 1152 `pert_p` rows, and every canonical, isomorphic and `pert_p2` row degenerated.
- **Torsion arm.** 48 distinct odd-order curves ($n \in [109, 93001]$) at precision p^4. The five torsion-section rates and the tau-ratio identity held on all 576 pairs (same-code evidence).
- **Non-canonical recoveries.** 0 successes in 3456 anomalous-degenerate rows and 0 in 576 rows per torsion section. The 95% one-sided bounds are 8.7×10^{-4} and 5.2×10^{-3} per row; these do not test the 1/p rate.

No claim is made about 33 to 96-bit fields, a deployed curve, or any attack cost.



EXP-CZLIFT-f5e06c; runs RUN-CZLIFT-032a92 (seed 20261009) and RUN-CZLIFT-6447f0 (seed 20261010); EV-CZLIFT-e9a7cf; DEC-20261023-9de83a. Toy tier (p ≤ 97813 < 2^17). A: one-sided Clopper-Pearson 95% bounds, 0 successes, rows = (curve lift, section, (P, d)); torsion bounds are per section (sections share (P, d) draws). B: from reports/redteam-20261023/rt_power_and_design.json.

Figure 1: Panel A shows one-sided 95% Clopper-Pearson upper bounds on the per-row exact-hit success rate (0 successes in every cell), against the recorded mean 1/p (the prediction) and the $p^{-1/2}$ range of the tested primes. The bars are drawn by `plot.py` from recorded p lists. The torsion bounds are per section, because the three sections share (P, d) draws. Panel B shows the red team's operating characteristic of the frozen $|z| \le 4$ clause. Rate 0 passes it with probability 1.000 and the predicted 1/p with probability 0.900. Runs RUN-CZLIFT-032a92 and RUN-CZLIFT-6447f0.

Observation

- **Validity.** Both planned runs are present: `completed_valid`, `output_validated`, seeds as frozen, `code.dirty: false`, 0 dropped rows. Manifest metrics equal the raw results (141/141 per seed), a full replay is byte-identical, and the blind re-derivation agrees (VAL-20261022-1d018a). Wall time was 10.85 s and 11.21 s.
- **Anomalous arm**, both seeds:
  - `rule_agreement` is 1.0 with 0 disagreements in 2304 rows per seed;
  - Smart ratio $= d$ on 576/576 nondegenerate rows per seed, all `pert_p`;
  - 1728 degenerate rows per seed with 0 next-digit successes, against 0.0591 and 0.0557 expected under 1/p;
  - recorded z is -0.2432 and -0.2361;
  - lift distances: $v_{p(j-j_{\mathrm{CM}})} = 1, 2, 8$ at $e = 1$ and $3, 6, 8$ at $e = 3$.
- **Torsion arm**, both seeds:

  - the five torsion rates are 1.0 on 288/288 pairs;
  - `identity_rate` is 1.0 in all 9 section × arm cells, with the non-vacuity control holding (`homomorphism_rate_prec2` 0.0);
  - 0/288 successes per section;
  - arm-difference z is null;
  - mean torsion lift cost is 107.17 and 107.75 complete additions, within the bound.
- **Driver sources**, read in this review:
  - the anomalous driver draws a fresh (P, d) per row (`EXP-CZLIFT-17b7eb/implementation/run.py`, lines 79-86);
  - the torsion driver reuses each (P, d) across the three sections (`EXP-CZLIFT-ae0a13/implementation/run.py`, lines 80-118);
  - both count points in $O(p)$ (`harness/toycurve.py`, lines 80-81).

Comparison

- **Frozen criteria.** The success criterion is met on both seeds; the falsification criterion is not.
- **Against the toy runs** (EXP-CZLIFT-17b7eb, $p \leq 1500$; EXP-CZLIFT-ae0a13, $p \leq 600$): the same exact outcomes recur on new curves with p up to 97813.
- **Rule disagreement.** With 0 disagreements, the 95% bound is 1.81% per curve × class cell.
- **Torsion failures.** With 0 failures, the 95% bound is 0.52% per pair.
- **1/p clauses.** The bounds are 1.73×10^{-3} per seed and 8.66×10^{-4} pooled on the anomalous arm (51× to 54× and 26× the mean 1/p). On the torsion arm they are 1.03×10^{-2} per seed and 5.19×10^{-3} pooled per section (352× to 395× and 187×).
  - P(all zero | 1/p) = 0.850, so the likelihood ratio of 1/p to rate 0 is 0.85.
  - The recorded z equals the design-only value $-\sqrt{\sum 1/p_i}$ to 1.2×10^{-15}.
  - One success on seed 20261010 would give $z = 4.0001$.
- **Against $p^{-1/2}$**, whose range over the tested primes is $[3.35, 9.24] \times 10^{-3}$ (anomalous) and $[3.20, 10.0] \times 10^{-3}$ (torsion):
  - the anomalous bounds lie below the whole range;
  - the pooled torsion per-section bound lies inside it.
- **Correction to the red team.** Its all-sections common-rate torsion bound (1.73×10^{-3}, $n = 1728$) assumes independent rows, which the driver does not provide, so it is not used.
- **Tier.** Fields of at most 17 bits are toy by the mechanical rule, so the tier clause cannot be met by this design.

Inference

- **Exact clauses.** At toy tier, with p up to 10^5, the exact clauses of H-CZLIFT-889c9d and H-CZLIFT-6d4ed1 hold with no exception on new curves, on two seeds. The anomalous valuations have independent-code corroboration; the torsion identities have same-code evidence only. The identities follow from stated derivations (Serre-Tate splitting, $\mathrm{Hom}(\mathbb{Z}/n, \mathbb{Z}_p) = 0$, Smart's theorem) whose mechanism does not depend on p. The run therefore adds instance coverage, not a new regime.
- **1/p clauses.** These are inconclusive: the data do not distinguish 1/p from 0. The anomalous bound excludes a $p^{-1/2}$ rate at every tested p; the torsion per-section bound alone does not.
- **Tier clause.** This fails on policy grounds, not because of a measurement, so no adverse transition is taken and no refutation artifact is needed.
- **Decision.** The decision is `refine`, not support or inconclusive.
  - Support would endorse a tier move the data cannot carry and an untested rate clause.
  - Inconclusive would discard a clean exact replication.
- **Earlier reading.** The reading in DEC-20261009-44c288 that success rates "stay at 1/p", and its "medium tier" boundary, are not adopted.

Limitation

- Toy tier only; no transfer argument to 33 to 96-bit fields or to a deployed curve.
- Same-code torsion replication.
- 41 of 84 anomalous curves sampled, not a census.
- The $e = 3$ threshold 6 is not distinguished from 4 or 5.
- Exact-hit counting cannot see partial leakage of d, because rows record no d or ratio.
- The structured small-d stratum is hit in about 1.5×10^{-4} of rows.
- The "1/p rate" is an unnumbered random-model assumption in all three hypotheses.
- The arm contrast confounds lift type with curve type.
- No memory figure is recorded and no package versions are pinned.

Next actions

1. **Ranked first; this is the goal's next action.** A numbered-heuristic validation contract, designed through `/design-experiment` from a refined hypothesis. The heuristic: the Smart (tau-) ratio of the `hensel_random`, `naive` and `teichmuller` sections is uniform on $\mathbb{F}_p$ and independent of d. The contract should:
   - use a new driver that records r and d per row;
   - pre-register residue-uniformity, independence and near-miss tests with tail checks;
   - add a structured-d stratum, a CM non-canonical-lift torsion arm, and the census of all 84 anomalous primes in $[10^4, 10^5]$;

- state the unit of analysis.

  Toy tier suffices for power. It is not approved here.
2. The separate reviews of H-CZLIFT-889c9d and H-CZLIFT-6d4ed1 should cite the EV-CZLIFT-e9a7cf clause readings: no medium tier, 1/p inconclusive, same-code torsion evidence, arm difference not evaluated.
3. Optional: an independent implementation of the torsion section, folded into the first action.
4. Deferred: the red team's 33 to 96-bit exact-clause successor. Its information value is low because the clauses are p -independent consequences of theorems. It also needs a new driver, because the lane-1 drivers count points in $O(p)$.

Reviews and links

- Validation (passed; J1, J2, J3 and PTM-1 hold): `reports/validation-20261022/validation_report.yaml`
- Red team (R1 breaks, R2 breaks, PTM-1 holds): `reports/redteam-20261023/redteam_report.yaml` and `rt_power_and_design.json`
- Records: `ledger/evidence/EV-CZLIFT-e9a7cf.yaml`, `ledger/decisions/DEC-20261023-9de83a.yaml`, `experiments/EXP-CZLIFT-f5e06c/analysis.md`
- Earlier run reports: none exist for this experiment (no `reports/run-*/` directory). Run records are under `experiments/EXP-CZLIFT-f5e06c/runs/`.
- Canonical graphs: none for this lane; `knowledge/frontiers/` has no CZLIFT graph, so nothing was refreshed.
- Graph source: `plot.py` reads `plot_data.json` (recorded values and red-team computations, with run IDs) and writes `graph.svg`.